The Cannonball Circus Caper

Whodunnit? · Quadratics · Teacher Guide and Answer Key

Unit	Time	Supplies	Group size
Quadratics	20–40 minutes	Pencil, scrap paper, scissors	Teams of 2–3 (or solo)

Skills practiced: Solving quadratics by factoring, square roots and the quadratic formula; vertex and axis of symmetry.

General procedures

1. Give each student the **student sheet** (story and case board) and the **Show Your Work** page.
2. Print the **Clue Cards** (pages 3–7 of the student packet) and cut them apart on the dashed lines. Laminating them lets you reuse the set. Post them around the room.
3. Students visit the cards in any order. At each card they solve the problem on their work page, find their answer among the four choices, read the clue beside it, and cross things off the case board. **Every wrong choice leads to a false clue** that clears the real culprit, so a mistake shows up when the board does not narrow down to one suspect, one place and one method.
4. When a team thinks it knows, they bring you the three names. If they are wrong, let them hunt for the card where something went wrong.

Solution: Bumbles took it, in the Cannon Platform, using this method: Fired from the cannon.

The reveal (read aloud): During the drumroll, **Bumbles the Clown** climbed onto the **Cannon Platform**, loaded the Golden Top Hat into the **cannon**, and fired it over the tent wall to a waiting wagon. Every alibi and method you cleared led to the clown. Ta-da!

Clue map: what each correct clue clears

Card	Clears
1	Suspect: Ringmaster Lopez · Place: Big Top Ring
2	Method: Squeezed in a clown car
3	Suspect: Zara · Place: Trapeze Platform
4	Method: Hidden in a popcorn tub
5	Suspect: Strongman Gus · Place: Popcorn Stand
6	Method: Rolled out in a barrel
7	Suspect: Madame Fortuna · Place: Animal Tent
8	Method: Stuffed in a drum
9	Suspect: Trainer Ines · Place: Clown Car Garage
10	Method: Carried in a juggler’s bag

Answer Key

The Cannonball Circus Caper · Correct choice and worked solution for each card

#	Problem	Correct choice	Worked solution	Clue it leads to
1	Solve by factoring: $x^2 - 5x + 6 = 0$	**D.** $x = 2, 3$	$(x - 2)(x - 3) = 0$ $x = 2, 3$	Ringmaster Lopez was announcing the next act in the Big Top Ring. The whole audience heard him.
2	Solve by factoring: $x^2 + 5x + 6 = 0$	**B.** $x = -2, -3$	$(x + 2)(x + 3) = 0$ $x = -2, -3$	The clown car was searched. Only 14 rubber chickens inside.
3	Solve by factoring: $x^2 + x - 6 = 0$	**B.** $x = 2, -3$	$(x - 2)(x + 3) = 0$ $x = 2, -3$	Zara was swinging on the Trapeze Platform with her safety line recorded on camera.
4	Solve by factoring: $x^2 - x - 6 = 0$	**D.** $x = -2, 3$	$(x + 2)(x - 3) = 0$ $x = -2, 3$	Every popcorn tub was emptied and counted. No hat.
5	Solve: $x^2 = 16$	**A.** $x = \pm 4$	take the square root of both sides $x = \pm 4$	Strongman Gus was lifting a pretzel for a kid at the Popcorn Stand. The vendor remembers.
6	Solve: $2x^2 = 50$	**C.** $x = \pm 5$	$x^2 = 25$ $x = \pm 5$	The barrel by the gate was rolled open. Just old costumes inside.
7	Use the quadratic formula to solve: $x^2 - 6x + 4 = 0$	**D.** $x = 3 \pm \sqrt{5}$	$a = 1, b = -6, c = 4$ $x = \frac{6 \pm \sqrt{(36 - 16)}}{2}$ $= \frac{6 \pm 2\sqrt{5}}{2}$ $x = 3 \pm \sqrt{5}$	Madame Fortuna was telling fortunes in the Animal Tent. Her appointment book shows it.
8	Find the vertex of $y = x^2 - 4x + 1$	**B.** $(2, -3)$	$x = \frac{-b}{2a} = \frac{4}{2} = 2$ $y = 2^2 - 4(2) + 1 = -3$ $(2, -3)$	The drum was opened. It only holds drumsticks.
9	Solve by factoring: $x^2 - 2x - 8 = 0$	**A.** $x = 4, -2$	$(x - 4)(x + 2) = 0$ $x = 4, -2$	Trainer Ines was washing the clown car in the Clown Car Garage with two helpers.
10	Find the equation of the axis of symmetry of $y = x^2 + 6x + 5$	**C.** $x = -3$	$x = \frac{-b}{2a} = \frac{-6}{2}$ $x = -3$	The juggler's bag was checked. Only bowling pins and one banana.

Wrong-Answer Map

The Cannonball Circus Caper · What a wrong choice looks like, and the false clue it leads to

Card	Wrong choice	Likely error	False clue
1	A. $x = -2, -3$	sign error on roots	Bumbles was honking his horn in the Big Top Ring. The front row can't stop laughing about it.
1	B. $x = 1, 6$	product 6 but sum 7	Zara was checking her helmet on the Cannon Platform before her act. A stagehand chatted with her the whole time.
1	C. $x = 5, 6$	used b and c as roots	The cannon's barrel was checked and is spotless. Nothing has been fired from it today.
2	A. $x = -1, -6$	product 6 but sum -7	Strongman Gus was hauling a heavy sandbag onto the Cannon Platform. Two stagehands counted his trips.
2	C. $x = 2, 3$	sign error on roots	Bumbles was juggling popcorn at the Popcorn Stand. The vendor has the buttery fingerprints to prove it.
2	D. $x = -5, -6$	used b and c as roots	The cannon's fuse is still wrapped in plastic. It was never lit.
3	A. $x = -1, 6$	product -6 but sum 5	The cannon's safety pin is locked in place. It could not have fired today.
3	C. $x = -2, 3$	sign error on roots	Bumbles was making balloon animals in the Animal Tent. The pony wore one as a hat.
3	D. $x = 1, -6$	product -6 but sum -5	Madame Fortuna was reading palms on the Cannon Platform. The cannon crew got very good fortunes.
4	A. $x = 2, -3$	sign error on roots	Bumbles was holding the net rope on the Trapeze Platform. The safety crew signed off on it.
4	B. $x = 1, -6$	product -6 but sum -5	The cannon crew swept the barrel clean and found nothing but one dusty piece of confetti.
4	C. $x = -1, 6$	product -6 but sum 5	Trainer Ines was calming a nervous pony on the Cannon Platform. The crowd below clapped for her.
5	B. $x = \pm 2$	took a fourth root	The cannon's powder tank is full to the brim. Not a drop was used.
5	C. $x = \pm 8$	divided by 2 instead of square root	Ringmaster Lopez was giving the safety speech on the Cannon Platform. The loudspeaker log has the time.
5	D. $x = 4$	forgot the negative root	Bumbles was squeezing into the tiny car in the Clown Car Garage for practice. Five clowns watched and counted.
6	A. $x = 5$	forgot $\pm$	Zara was tying her sparkly boots on the Cannon Platform. The photographer snapped a picture of it.
6	B. $x = \pm\sqrt{50}$	forgot to divide by 2	The net below the cannon is dry and unmarked. Nobody landed in it today.
6	D. $x = \pm 25$	forgot the square root	Bumbles was tripping over his big shoes in the Big Top Ring. The audience gasped and laughed.
7	A. $x = -3 \pm \sqrt{5}$	sign error on $-b$	Bumbles was squirting seltzer at the Popcorn Stand. The vendor still has a wet apron.
7	B. $x = 3 \pm \sqrt{10}$	divided under the radical by 2	The cannon's launch counter reads zero. It hasn't gone off since last week's show.
7	C. $x = 3 \pm 2\sqrt{5}$	forgot to divide the radical by 2	Strongman Gus was tightening bolts on the Cannon Platform. The crew chief signed his work order.
8	A. (4, 1)	used $x = \frac{-b}{a}$	The cannon crew swears it stayed covered all evening. The cover's zipper is still sealed.

8	C. (−2, 13)	sign error in $x = \frac{-b}{2a}$	Bumbles was waving from the Trapeze Platform in his rainbow wig. The crowd below spotted him right away.
8	D. (2, 1)	used c as y-coordinate	Madame Fortuna was waving her scarf at the Cannon Platform to bless the show. A dozen people watched her.
9	B. x = 8, −1	product −8 but sum 7	Ringmaster Lopez was testing the spotlight from the Cannon Platform. The lighting crew confirmed it.
9	C. x = −4, 2	sign error on roots	Bumbles was teaching a parrot to honk in the Animal Tent. The trainer's notebook mentions it.
9	D. x = 4, 2	dropped sign of root	Soot marks inside the cannon are old and faded. Nothing has been fired lately.
10	A. x = −6	used −b instead of $\frac{-b}{2a}$	Bumbles was repainting the clown car's polka dots in the Clown Car Garage. His paint-covered shoes are still there.
10	B. x = −5	used c	A stagehand checked the cannon with a flashlight. It was empty from end to end.
10	D. x = 3	sign error	Trainer Ines was posing with a tiger cub on the Cannon Platform. A kid's photo shows the time.